Dataset Description

1. Metadata_V2.csv

The file **Metadata_V2.csv** contains data on the collected samples.

Column name	Explanation
Lab_ID	Unique sample identifier in the format F-farm_number-sample_number (e.g., F1_001).
Sampling_status	Indicates whether the sample was collected (Yes / No).
Sampling	Corresponds to the sampling period (1 or 2).
Farm	Farm number.
Cow	Cow identifier, consisting of the farm number and a letter representing the cow (e.g., 1A).
Date	Sampling date.
Matrices	Sample type: Feces /Milk /Bedding /Milk_filter / Tank_milk.
Parity	Number of lactations of the cow.
Breed	Cow breed.
Calving_date	Date of the last calving before sampling.
qPCR_date	Date of qPCR analysis.
qPCR_ct_value	Cycle threshold (Ct) value from qPCR.
qPCR_unspecific	Non-specific amplification detected in qPCR.
qPCR_log16S	Intermediate calculation for 16S qPCR: Log of the number of 16S DNA copies per well after modification of the CT data with ranges that have been carried out for each assay plate.
qPCR_16Spuit	Intermediate calculation for qPCR 16S : DNA 16S copy number per well.
qPCR_16Sech	Number of 16S DNA copies per sample, considering the variable assay between matrices.
Somatic_cells	Number of somatic cells counted in milk.
DNA_extraction_date	Date of DNA extraction.
Antibiotic	Indicates whether the cow received antibiotic treatment (Yes / No).
Antibiotic_administration	Route of antibiotic administration: Intramammary /Other / Intramammary_Other (for both routes used).
Antibiotic_family	Family of antibiotic used

2. qPCR_range.csv

The file **qPCR_range.csv** contains qPCR standard curve data used for quantifying pathogen levels in the samples. This dataset includes two types of standard curves:

1. Standards derived from the positive controls of the PathoProof kit.
2. Standards for two Staphylococcus species: *Staphylococcus aureus* and *Staphylococcus xylosus*.

Column name	Explanation
Sample	Sample used for ranges: Positive_control / Staph_aureus / Staph_xylosus.
Plate_ID	Plate identifier: Calibration_1 / Calibration_staph.
PCR_date	Date of PCR analysis.
Dilution_type	Dilution factor used.
Dilution_number	Number of dilutions.
Lab_ID	Unique sample identifier.
Pathogen	List of 15 pathogens and the beta-lactamase gene detected by PathoProof kit.
Quantity	Categories of bacterial presence depend on the Ct value.
Proportion	When multiple bacterial targets are detected in a sample, the software reports the percentage of the most abundant bacterial species, if its proportion is over 90%.
CT_value	Number of cycles supplied by the kit.
Control	Warning.
Select_calibration	Is this sample kept for the calculation of the range? Some samples are not kept.

3. qPCR_pathogen.csv

The file **qPCR_pathogen.csv** contains PathoProof kit data for all samples, including both positive and negative controls.

Column name	Explanation
Lab_ID	Unique sample identifier in the format F-farm_number-sample_number (e.g., F1_001).
Type_sample	Sample / Negative_PCR_control / Positive_PCR_control / Negative_extraction_control / Positive_extraction_control.
Plate_ID	Plate identifier.
PCR_date	Date of PCR analysis.
Pathogen_...	Quantity / proportion / CT_value / control data for all pathogens and the antibiotic resistance gene
..._quantity	Categories of bacterial presence depend on the Ct value.
..._proportion	When multiple bacterial targets are detected in a sample, the software reports the percentage of the most abundant bacterial species, if its proportion is over 90%.
..._CT_value	Number of cycles supplied by the kit.
..._control	Warning.

4. Missing_sampling.csv

File containing dates for each farm where sampling was not carried out. This file is used at the start of the qPCR_pathogen_analysis.R script to adjust the gaps between samples to 3 or 4 days to ensure consistency.

5. Data Processing & Analysis

- The dataset **qPCR_range.csv** is used to calculate **standard curve parameters** and transform **qPCR_pathogen.csv** data using the **qPCR_pathogen_preprocessing.R** script.
- The processed data is then analysed in **qPCR_pathogen_analysis_V2.R**, generating the figures included in the associated research article.
- Two **Python scripts** are used for additional visualization:
    - **Transitions_graph_V2.py**: Generates transition matrices.
    - **Transition_antibiotic_V2.py**: Analyses transitions associated with antibiotic treatments.
- These Python scripts require datasets generated by **qPCR_pathogen_analysis.R**.